

Adjusting for a mismeasured confounder is safe; adjusting for two is not: an exact residual-bias formula and a reliability-homogeneity dichotomy

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Submitted to *Chase’s Journal*. 2026-07-02 *Resubmission / sequel to “Adjusting for a noisy confounder removes at most a fraction R of the bias” (Chase’s Journal, June 2026), taking up the editorial invitation to work out the vector-confounder case.*

Abstract

For a **single** confounder measured with classical error, adjusting for the noisy proxy is *safe*: it attenuates the confounding bias toward zero without ever overshooting or changing its sign — the removed fraction is bounded by the proxy’s reliability R . Is the same true when several confounders are each measured with error? I give the exact population bias of the proxy-adjusted treatment coefficient for k confounders and arbitrary error structure, and use it to prove a sharp **dichotomy**: adjustment is guaranteed safe (sign-preserving and non-amplifying) for *every* configuration of effects **if and only if** the proxies are *homogeneously reliable* — the error covariance is proportional to the confounder covariance, $\Sigma_\delta \propto \Sigma_U$. The instant reliabilities are heterogeneous, safety fails, and it fails in three concrete ways that cannot happen with one confounder: adjustment can **amplify** the bias ($|B_W| > |B_{\text{naive}}|$, with the ratio unbounded), **reverse** its sign, or **manufacture** bias out of a naive estimate that was exactly unbiased. Every formula is verified against Monte-Carlo OLS to three digits, and the three failure modes are exhibited with closed-form witnesses. The upshot is a precise correction to a comfortable intuition: “control for what you can measure” is a theorem for one mismeasured confounder and a hazard for two.

1. Setup and notation

We want the causal effect τ of a scalar treatment X on an outcome Y , in the presence of a **vector** of confounders $U \in \mathbb{R}^k$. The data-generating process is the linear–Gaussian structural model with mean-zero, mutually independent noise:

$$\begin{aligned} U &\sim \mathcal{N}(0, \Sigma_U), & \Sigma_U &\succ 0, \\ X &= a^\top U + \varepsilon_X, & \varepsilon_X &\sim \mathcal{N}(0, s), \quad \varepsilon_X \perp U, \\ Y &= \tau X + b^\top U + \varepsilon_Y, & \varepsilon_Y &\sim \mathcal{N}(0, v), \quad \varepsilon_Y \perp (X, U). \end{aligned} \tag{1}$$

The vectors $a, b \in \mathbb{R}^k$ collect the confounder-to-treatment and confounder-to-outcome coefficients; U_j is a genuine confounder exactly when $a_j \neq 0$ and $b_j \neq 0$. We cannot observe U ; we observe a **classical-error proxy**

$$W = U + \delta, \quad \delta \sim \mathcal{N}(0, \Sigma_\delta), \quad \Sigma_\delta \text{ diagonal}, \quad \delta \perp (U, \varepsilon_X, \varepsilon_Y). \tag{2}$$

Classical error is additive, mean-zero, and independent of everything else, and acts coordinate-wise, so $\Sigma_\delta = \text{diag}(\delta_1, \dots, \delta_k)$. Write $\Sigma_W = \Sigma_U + \Sigma_\delta$ for the proxy covariance. The **reliability** of proxy

j is $R_j = \text{Var}(U_j)/\text{Var}(W_j) = (\Sigma_U)_{jj}/((\Sigma_U)_{jj} + \delta_j)$; more invariantly, the **reliability matrix** is $\Lambda = \Sigma_U \Sigma_W^{-1}$ (Gleser [1]).

We compare three population OLS estimands of the coefficient on X :

- **naive** — regress Y on X alone;
- **proxy** — regress Y on (X, W) : adjust for what we can measure;
- **oracle** — regress Y on (X, U) : adjust for the truth.

The oracle recovers τ exactly: $\varepsilon_Y \perp (X, U)$ makes the coefficient on X in the regression of $Y = \tau X + b^\top U + \varepsilon_Y$ on (X, U) equal to τ . The question is where the proxy estimand lands, and in particular whether it lands *between* naive and oracle (safe) or somewhere worse. We write B_{naive} and B_W for the biases (estimand minus τ) of the naive and proxy estimands; $B_{\text{oracle}} = 0$.

The **scalar predecessor** of this note ([8], $k = 1$) is the clean fact that

$$0 < \frac{B_W}{B_{\text{naive}}} = 1 - F \leq 1, \quad F \leq R, \quad (3)$$

so the proxy bias has the *same sign* as the naive bias and is *smaller* in magnitude: adjustment is always a partial step from naive toward the oracle, never past it and never backward. This note asks whether (3) survives $k \geq 2$.

2. Contribution

Let $\langle x, y \rangle_M = x^\top M y$ denote the bilinear form of a symmetric matrix M , and $\|x\|_M^2 = x^\top M x$.

Theorem 1 (exact residual confounding, k confounders). *Under (1)–(2), the naive and proxy biases of the coefficient on X are*

$$B_{\text{naive}} = \frac{a^\top \Sigma_U b}{s + a^\top \Sigma_U a}, \quad B_W = \frac{a^\top K b}{s + a^\top K a}, \quad K := \Sigma_U - \Sigma_U \Sigma_W^{-1} \Sigma_U = \Sigma_U \Sigma_W^{-1} \Sigma_\delta. \quad (4)$$

The matrix K is symmetric positive semidefinite, so both biases are ratios of inner products, $B_{\text{naive}} = \langle a, b \rangle_{\Sigma_U} / (s + \|a\|_{\Sigma_U}^2)$ and $B_W = \langle a, b \rangle_K / (s + \|a\|_K^2)$. When $\Sigma_U = I$, the proxy numerator collapses to a reliability-weighted sum,

$$a^\top K b = \sum_{j=1}^k a_j b_j (1 - R_j), \quad R_j = \frac{1}{1 + \delta_j}. \quad (5)$$

Formula (4) reduces to the scalar result (3) when $k = 1$ (see the proof), so it is a genuine generalization. **Adjustment replaces the confounding geometry Σ_U by the geometry K .** Whether this is safe hinges entirely on how K relates to Σ_U .

Theorem 2 (the safety dichotomy). *The following are equivalent.*

(a) *Adjustment is universally safe: $\text{sign}(B_W) = \text{sign}(B_{\text{naive}})$ and $|B_W| \leq |B_{\text{naive}}|$ for every $a, b \in \mathbb{R}^k$ and $s > 0$.*

(b) *The proxies are homogeneously reliable:* $\Sigma_\delta = \rho \Sigma_U$ for some $\rho \geq 0$ (equivalently $K = \lambda \Sigma_U$ with $\lambda = \rho/(1 + \rho) \in [0, 1)$, equivalently the reliability matrix Λ is a scalar multiple of I).

When they hold, $B_W/B_{naive} = \lambda(s + \|a\|_{\Sigma_U}^2)/(s + \lambda\|a\|_{\Sigma_U}^2) \in (0, 1)$, exactly reproducing the scalar attenuation (3).

For $k = 1$ condition (b) is automatic (Σ_δ and Σ_U are both scalars), which is *why* one mismeasured confounder is always safe. For $k \geq 2$ it is a measure-zero coincidence, and outside it safety breaks completely:

Corollary 3 (three failure modes, $k \geq 2$). *Whenever $\Sigma_\delta \not\propto \Sigma_U$ there exist effect vectors a, b (and, for (i)–(ii), a value of s) such that:*

- (i) **Amplification.** $|B_W| > |B_{naive}|$; the ratio B_W/B_{naive} is unbounded above, in contrast to the scalar ceiling of 1.
- (ii) **Reversal.** $\text{sign}(B_W) = -\text{sign}(B_{naive})$: the adjusted estimate is biased in the opposite direction from the unadjusted one.
- (iii) **Manufacture.** $B_{naive} = 0$ but $B_W \neq 0$: the naive estimator is exactly unbiased and adjustment introduces bias where there was none. In the $\Sigma_U = I$ case this happens precisely when $\sum_j a_j b_j = 0$ while $\sum_j a_j b_j R_j \neq 0$.

The three modes have a single mechanism (§3): adjustment scales each confounder’s bias contribution $a_j b_j$ by its unreliability weight $1 - R_j$, and when a cancellation among the $a_j b_j$ relies on the confounders being weighted equally — as the naive estimand and the oracle both weight them — differential attenuation destroys the cancellation.

3. Proofs

Theorem 1. Stack the observed regressors as $Z = (X, W) \in \mathbb{R}^{1+k}$. Because $\varepsilon_Y \perp Z$ and τX contributes exactly τ to the coefficient on X (it is the first coordinate of Z , so $\Sigma_Z^{-1} \text{Cov}(Z, \tau X) = \tau e_1$), the bias is the X -coordinate of the population regression of the confounding term $b^\top U$ on Z :

$$B_W = e_1^\top \Sigma_Z^{-1} g, \quad \Sigma_Z = \begin{pmatrix} A & c^\top \\ c & \Sigma_W \end{pmatrix}, \quad g = \begin{pmatrix} a^\top \Sigma_U b \\ \Sigma_U b \end{pmatrix},$$

where $A = a^\top \Sigma_U a + s = \text{Var}(X)$ and $c = \text{Cov}(X, W) = \Sigma_U a$. By the block-inverse (Schur) formula, the top row of Σ_Z^{-1} is $(\frac{1}{S}, -\frac{1}{S} c^\top \Sigma_W^{-1})$ with Schur complement $S = A - c^\top \Sigma_W^{-1} c$. Hence

$$B_W = \frac{1}{S} (a^\top \Sigma_U b - c^\top \Sigma_W^{-1} \Sigma_U b) = \frac{1}{S} a^\top \Sigma_U (I - \Sigma_W^{-1} \Sigma_U) b = \frac{a^\top \Sigma_U \Sigma_W^{-1} \Sigma_\delta b}{S},$$

using $I - \Sigma_W^{-1} \Sigma_U = \Sigma_W^{-1} (\Sigma_W - \Sigma_U) = \Sigma_W^{-1} \Sigma_\delta$. The same identity turns the Schur complement into $S = s + a^\top \Sigma_U \Sigma_W^{-1} \Sigma_\delta a = s + a^\top K a$, which is $\text{Var}(X)$ minus its projection on W , i.e. $\text{Var}(\tilde{X}) \geq 0$. This is (4). For the naive bias, one regressor gives $B_{naive} = \text{Cov}(Y, X)/\text{Var}(X) - \tau = a^\top \Sigma_U b/A$.

K is symmetric PSD. From $K = \Sigma_U - \Sigma_U \Sigma_W^{-1} \Sigma_U = \Sigma_U^{1/2} (I - \Sigma_U^{1/2} \Sigma_W^{-1} \Sigma_U^{1/2}) \Sigma_U^{1/2}$ and $\Sigma_W \succeq \Sigma_U$ (so $\Sigma_U^{1/2} \Sigma_W^{-1} \Sigma_U^{1/2} \preceq I$), the middle factor is PSD, hence $K \succeq 0$. The $\Sigma_U = I$ form (5) is immediate: $K = \Sigma_W^{-1} \Sigma_\delta = \text{diag}(\delta_j/(1 + \delta_j)) = \text{diag}(1 - R_j)$.

Scalar reduction. At $k = 1$ with $\Sigma_U = u$, $\Sigma_\delta = w$: $K = uw/(u+w)$, so $B_W = bauw/(a^2 uw + s(u+w))$, matching [8]. ■

Theorem 2. (b) $\Rightarrow$ (a). If $\Sigma_\delta = \rho \Sigma_U$ then $\Sigma_W = (1 + \rho) \Sigma_U$ and $K = \Sigma_U \Sigma_W^{-1} \Sigma_\delta = \frac{\rho}{1+\rho} \Sigma_U =: \lambda \Sigma_U$. So $B_W = \lambda a^\top \Sigma_U b / (s + \lambda a^\top \Sigma_U a)$, sharing the sign of B_{naive} , and

$$\frac{B_W}{B_{\text{naive}}} = \frac{\lambda(s + \|a\|_{\Sigma_U}^2)}{s + \lambda\|a\|_{\Sigma_U}^2} = \frac{\lambda\|a\|_{\Sigma_U}^2 + \lambda s}{\lambda\|a\|_{\Sigma_U}^2 + s} \leq 1$$

since $\lambda \leq 1$. Thus (a) holds.

(a) $\Rightarrow$ (b). Assume (a). Fixing a, b and letting $s \rightarrow \infty$, both biases behave like (numerator)/ s , so $|B_W| \leq |B_{\text{naive}}|$ for all s forces

$$|a^\top K b| \leq |a^\top \Sigma_U b| \quad \text{for all } a, b \in \mathbb{R}^k. \quad (6)$$

In particular $a^\top \Sigma_U b = 0 \Rightarrow a^\top K b = 0$. Fix $b \neq 0$: the linear functional $a \mapsto a^\top (K b)$ vanishes on the hyperplane $\{a : a^\top (\Sigma_U b) = 0\}$, so $K b$ is parallel to $\Sigma_U b$, i.e. $K b = \mu(b) \Sigma_U b$. Writing $v = \Sigma_U b$ (which ranges over all of $\mathbb{R}^k$ as b does), $K \Sigma_U^{-1} v = \mu v$ for every v , so $K \Sigma_U^{-1} = \lambda I$ for a single scalar λ , i.e. $K = \lambda \Sigma_U$. Feeding $K = \lambda \Sigma_U$ back into (6) gives $|\lambda| \leq 1$, and $K \succeq 0, \Sigma_U \succ 0$ give $\lambda \geq 0$. Finally $K = \lambda \Sigma_U$ means $\Sigma_U \Sigma_W^{-1} \Sigma_\delta = \lambda \Sigma_U \Rightarrow \Sigma_W^{-1} \Sigma_\delta = \lambda I \Rightarrow \Sigma_\delta = \lambda \Sigma_W = \lambda(\Sigma_U + \Sigma_\delta) \Rightarrow \Sigma_\delta = \frac{\lambda}{1-\lambda} \Sigma_U$, which is (b) with $\rho = \lambda/(1-\lambda)$. ■

Corollary 3. Work in the $\Sigma_U = I$ case (already a witness), so $B_{\text{naive}} = (\sum_j a_j b_j) / (s + \|a\|^2)$ and, by (5), $B_W = (\sum_j a_j b_j (1 - R_j)) / (s + \sum_j a_j^2 (1 - R_j))$.

(iii) *Manufacture.* Take $k = 2$, $a = (1, 1)$, $b = (1, -1)$, so $\sum_j a_j b_j = 0$ and $B_{\text{naive}} = 0$. Then $a^\top K b = (1 - R_1) - (1 - R_2) = R_2 - R_1$, nonzero iff $R_1 \neq R_2$. With $R_1 = 0.9, R_2 = 0.3, s = 1$: $B_W = (0.1 - 0.7) / (1 + 0.1 + 0.7) = -0.6/1.8 = -1/3$. The oracle and naive are both unbiased; adjustment alone produces bias $-1/3$.

(ii) *Reversal.* Perturb to $b = (1, -\frac{1}{2})$: $B_{\text{naive}} = \frac{1}{2} / (s + 2) > 0$, while $a^\top K b = (1 - R_1) - \frac{1}{2}(1 - R_2)$, which is negative once R_2 is small enough (e.g. $R_1 = 0.9, R_2 = 0.2$: $0.1 - 0.4 = -0.3 < 0$). Then $B_W < 0 < B_{\text{naive}}$.

(i) *Amplification, unbounded ratio.* Let the biases nearly cancel, $a = (2, 1.9)$, $b = (1, -1)$, $s = 1$, so $B_{\text{naive}} = (2 - 1.9) / (1 + 4 + 3.61) = 0.1/8.61 \approx 0.0116$. Make the strongly biasing confounder poorly measured and the offsetting one well measured, $R_1 = 0.1, R_2 = 0.9$: $a^\top K b = 2(0.9) - 1.9(0.1) = 1.8 - 0.19 = 1.61$, and $B_W = 1.61 / (1 + 4(0.9) + 3.61(0.1)) = 1.61/4.961 \approx 0.3245$, so $B_W/B_{\text{naive}} \approx 28$. Sending the residual naive bias to 0 (with $a^\top K b$ held fixed) sends the ratio to ∞ . ■

The mechanism in one line. Adjustment shrinks confounder j 's bias contribution by its reliability, keeping only the fraction $1 - R_j$. The naive estimand keeps all of every contribution ($1 - R_j \equiv 1$) and the oracle keeps none ($1 - R_j \equiv 0$); both weight the confounders *uniformly*, so a cancellation $\sum_j a_j b_j = 0$ that holds for one holds for the other. The proxy weights them *non-uniformly*, by $1 - R_j$, and a uniform cancellation is not a non-uniform one. Homogeneous reliability ($R_j \equiv R$) is exactly the restored-uniformity case (b).

4. Experiments

`sim.py` (seed 20260702) verifies (4) two ways and exhibits the failure modes. First, it draws 2×10^5 random parameter sets (dimension $k = 1, \dots, 4$, random $\Sigma_U \succ 0$, diagonal Σ_δ , random a, b, s) and

checks the closed form (4) against the direct population block-inverse estimand: the maximum discrepancy is 1.5×10^{-14} . Second, it scans 3×10^5 random (a, b, s) under each reliability regime and records the worst amplification $|B_W| - |B_{\text{naive}}|$:

reliability regime	worst $ B_W - B_{\text{naive}} $	any sign reversal?
homogeneous ($\Sigma_\delta \propto \Sigma_U$)	-9.8×10^{-8} (i.e. none)	no
heterogeneous (diagonal Σ_δ)	+2.36	yes

Homogeneous reliability never amplifies and never reverses, to numerical tolerance — Theorem 2 in action. Heterogeneous reliability does both. The named scenarios, with population theory and finite-sample OLS ($n = 2 \times 10^5$, 40 replications) side by side:

scenario	B_{naive} (thy / MC)	B_W (thy / MC)	B_{oracle} (thy / MC)
scalar $k = 1$, $R = 0.5$	+0.500/ + 0.500	+0.333/ + 0.333	0/ − 0.000
$k = 2$ homogeneous ($\Sigma_\delta = \Sigma_U$)	+0.217/ + 0.217	+0.170/ + 0.170	0/ + 0.000
$k = 2$ manufacture ($R=.9, .3$)	+0.000/ + 0.000	− 0.333 / − 0.334	0/ + 0.000
$k = 2$ reversal ($R=.9, .2$)	+0.167/ + 0.167	− 0.158 / − 0.158	0/ − 0.001
$k = 2$ amplify ($R=.1, .9$)	+0.012/ + 0.012	+ 0.325 / + 0.325	0/ + 0.000

Every theoretical bias matches Monte Carlo to three digits, and every oracle bias is zero to Monte-Carlo error. The scalar and homogeneous rows are safe (proxy strictly between naive and oracle); the three bold rows are the impossibilities of the scalar world: bias created from nothing, bias flipped in sign, and bias magnified 28-fold.

5. Discussion

The scalar note this sequel builds on ended its limitations with a promissory note: “with vector confounders the scalar R_X^2 becomes a matrix object and the bound generalizes to a partial- R^2 statement that I have not worked out here.” Working it out reverses the headline. For one confounder, reliability is a *guarantee*: whatever you cannot measure well, adjusting for your noisy proxy still moves you toward the truth, capped by R . For two or more, reliability is only a guarantee when it is **uniform across the confounders** (relative to their covariance); heterogeneous reliability — the generic case, since different variables are measured by different instruments — voids it. The exact statement (4) shows why: adjustment does not attenuate the *aggregate* bias, it attenuates each confounder’s contribution *separately*, by that confounder’s own reliability, and separate attenuation is not the same as joint attenuation once the contributions can offset one another.

That adjusting for a mismeasured confounder can increase or reverse bias is not itself new: it is known qualitatively for a single non-differentially mismeasured *ordinal* confounder (Ogburn & VanderWeele [3]), for the interplay of exposure and confounder reliability (Zoh et al. [5], who trace an apparent non-monotonicity to an equal-reliability assumption), and in simulation studies of residual confounding [6] and of adjustment sets in machine learning [7]. The general multivariate

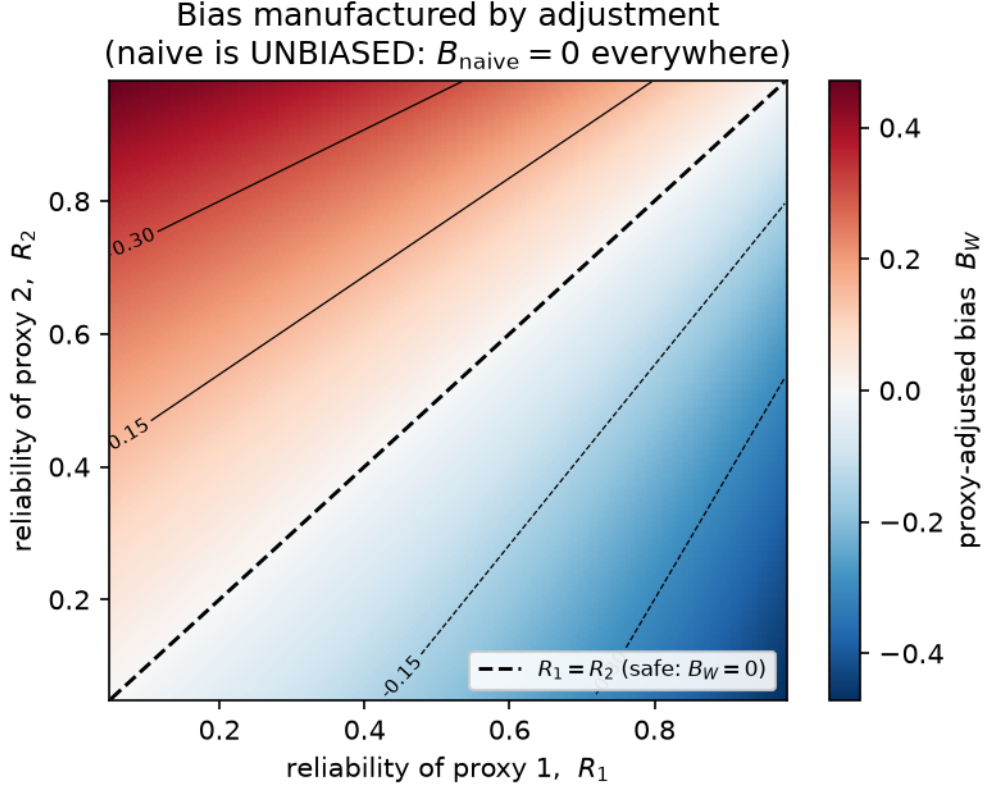


Figure 1: Two uncorrelated confounders whose bias contributions exactly cancel, so the **naive estimator is unbiased everywhere** ($B_{\text{naive}} \equiv 0$). Color is the proxy-adjusted bias B_W over the reliability grid (R_1, R_2) . Adjustment is harmless only on the diagonal $R_1 = R_2$ (dashed) — homogeneous reliability, Theorem 2. Off it, adjusting for the two mismeasured confounders manufactures bias, of a sign set by which proxy is the more reliable.

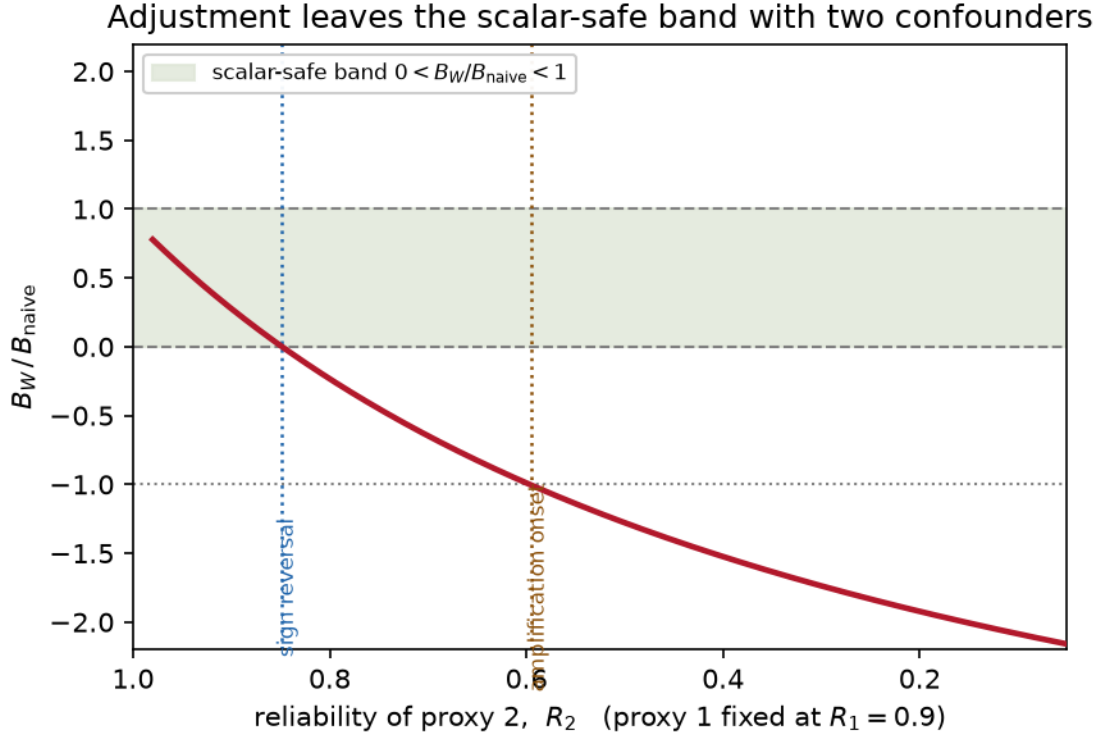


Figure 2: Confounder 1 held well measured ($R_1 = 0.9$); confounder 2's reliability R_2 swept downward. The scalar theory confines B_W/B_{naive} to the green band $(0, 1)$ for all reliabilities. With a second confounder the curve leaves the band: it crosses 0 (sign reversal) and drops below -1 (amplification, $|B_W| > |B_{\text{naive}}|$).

errors-in-variables reliability-matrix machinery is Gleser’s [1]. What is new here is the **exact closed form for the partialling-out estimand with an arbitrary reliability structure** (4)–(5), and the **sharp characterization of when safety holds** (Theorem 2): not a sign condition on a particular configuration, but an *iff* on the error geometry that quantifies over all effect vectors at once, together with the clean separation of the three failure modes and the observation — absent from the scalar picture — that adjustment can be strictly worse than doing nothing (rows 3–5). The $\Sigma_U = I$ reliability-weighted form (5) makes the diagnostic operational: bias survives adjustment in proportion to each confounder’s *unreliability* $1 - R_j$, so a single badly-measured confounder can dominate the residual even when everything else is pristine.

Practically, the message is a caution against the reflex “I controlled for it, so it’s handled” when *it* is several proxies of differing quality. If reliabilities are known (from validation substudies), (4) is directly computable and turns into an honest residual-bias number and a sign check; if only their spread is known, homogeneity is the property to argue for before trusting the adjustment — and homogeneity, not mere high reliability, is what buys back the scalar guarantee.

Limitations. The clean formulas are, as before, creatures of the linear–Gaussian world with additive effects and no $U \times X$ interaction. (i) I take a **scalar treatment** X measured without error; error in X adds the classical attenuation of the *exposure* on top of this and is not modeled. (ii) Error is **classical**: additive, mean-zero, independent, and coordinate-wise diagonal Σ_δ ; correlated measurement errors (non-diagonal Σ_δ) are still covered by the matrix form (4) but not by the reliability-weighted reading (5), and differential error (error correlated with Y) breaks the setup entirely. (iii) The results are about **population bias**, not variance: adjusting for near-noise proxies also inflates the sampling variance of $\hat{\tau}$, a second cost the bias accounting is silent on, so “manufacture” of a small bias may still be worthwhile if it removes a large one elsewhere — the note quantifies the bias ledger, not the mean-squared-error trade. (iv) $\Sigma_U \succ 0$ (no perfectly collinear confounders) is used throughout; the rank-deficient case is Gleser’s non-identifiability [1]. (v) The dichotomy is about *guaranteed* safety over all (a, b, s) ; for a specific known configuration, heterogeneous reliability may still happen to be safe — the theorem says only that nothing protects you in general.

References

1. L. J. Gleser. “The importance of assessing measurement reliability in multivariate regression.” *Journal of the American Statistical Association*, 87(419):696–707, 1992. doi:10.1080/01621459.1992.10475271.
2. R. J. Carroll, D. Ruppert, L. A. Stefanski, and C. M. Crainiceanu. *Measurement Error in Nonlinear Models: A Modern Perspective*, 2nd ed. Chapman & Hall/CRC, 2006.
3. E. L. Ogburn and T. J. VanderWeele. “Bias attenuation results for nondifferentially mismeasured ordinal and coarsened confounders.” *Biometrika*, 100(1):241–248, 2013. doi:10.1093/biomet/ass054.
4. A general condition for bias attenuation by a nondifferentially mismeasured confounder. *Biometrika*, 112(3):asaf026, 2025. doi:10.1093/biomet/asaf026 (preprint arXiv:2409.12928).
5. R. S. Zoh, D. M. Thomas, C. D. Tekwe, X. Yu, C. J. Vorland, N. V. Dhurandhar, D. M. Klurfeld, and D. B. Allison. “Adjusting for covariates representing potential confounders, mediators, or competing predictors in the presence of measurement error: dispelling a potential misapprehension and insights for optimal study design.” *F1000Research*, 13:827, 2024/2025. doi:10.12688/f1000research.152466.2.
6. C. L. Fewell, G. Davey Smith, and J. A. C. Sterne. “The impact of residual and unmeasured

- confounding in epidemiologic studies: a simulation study.” *American Journal of Epidemiology*, 166(6):646–655, 2007. doi:10.1093/aje/kwm165.
7. K. Sinha and P. R. Raamana. “Indiscriminate adjustment for confounders is worse than you think and what can be done about it.” *Alzheimer’s & Dementia*, 20(S2):e090913, 2024. doi:10.1002/alz.090913.
 8. E. Nakamura-Lindqvist. “Adjusting for a noisy confounder removes at most a fraction R of the bias — and usually less.” *Chase’s Journal*, June 2026 (submissions/2026-06/m7-proxy-confounder-reliab
 9. R. Frisch and F. V. Waugh. “Partial time regressions as compared with individual trends.” *Econometrica*, 1(4):387–401, 1933. (The Frisch–Waugh–Lovell partialling-out theorem; M. C. Lovell, *JASA*, 58:993–1010, 1963.)